

Chapter 1 Vector Spaces (4 hours)

1.A $\mathbb{R}^n$ and $\mathbb{C}^n$: Complex Numbers, Lists, $\mathbb{F}^n$, Digression on Fields

1.B Definition of Vector Space

1.C Subspaces: Sums of Subspaces, Direct Sums

第一章 向量空间 (4 小时)

1.A $\mathbb{R}^n$ 与 $\mathbb{C}^n$

1.B 向量空间的定义

1.C 子空间

Chapter 2 Finite-Dimensional Vector Spaces (4 hours)

2.A Span and Linear Independence: Linear Combinations and Span, Linear Independence.

2.B Bases

2.C Dimension

第二章 有限维向量空间 (4 小时)

2.A 张成空间与线性无关

2.B 基

2.C 维数

Chapter 3 Linear Maps (10 hours)

3.A The Vector Space of Linear Maps: Definitions and Examples of Linear Maps, Algebraic Operations on $L(V, W)$.

3.B Null Spaces and Ranges: Null Space and Injectivity, Range and Surjectivity, Fundamental Theorem of Linear Maps.

3.C Matrices: Representing a Linear Map by a Matrix, Addition and Scalar Multiplication of Matrices, Matrix Multiplication.

3.D Invertibility and Isomorphic Vector Spaces: Invertible Linear Maps, Isomorphic Vector Spaces, Linear Maps Thought of as Matrix Multiplication, Operators.

3.E Products and Quotients of Vector Spaces: Products of Vector Spaces, Products and Direct Sums, Quotients of Vector Spaces.

3.F Duality: The Dual Space and the Dual Map, The Null Space and Range of the Dual of a Linear Map, The Matrix of the Dual of a Linear Map, The Rank of a Matrix.

第三章 线性映射 (10 小时)

3.A 向量空间的线性映射

3.B 零空间与值域

3.C 矩阵

3.D 可逆性与同构的向量空间

3.E 向量空间的积与商

3.F 对偶

Chapter 4 Polynomials (4 hours)

Complex Conjugate and Absolute Value, Uniqueness of Coefficients for Polynomials, The Division Algorithm for Polynomials, Zeros of Polynomials, Factorization of Polynomials over $\mathbb{C}$, Factorization of Polynomials over $\mathbb{R}$.

第四章 多项式 (4 小时)

Chapter 5 Eigenvalues, Eigenvectors, and Invariant Subspaces (6 hours)

5.A Invariant Subspaces: Eigenvalues and Eigenvectors, Restriction and Quotient Operators.

5.B Eigenvectors and Upper-Triangular Matrices: Polynomials Applied to Operators, Existence of Eigenvalues, Upper-Triangular Matrices.

5.C Eigenspaces and Diagonal Matrices

第五章 本征值、本征向量、不变子空间 (6 小时)

5.A 不变子空间

5.B 本征向量与上三角矩阵

5.C 本征空间与对角矩阵

Chapter 6 Inner Product Spaces (6 hours)

6.A Inner Products and Norms: Inner Products, Norms.

6.B Orthonormal Bases: Linear Functionals on Inner Product Spaces.

6.C Orthogonal Complements and Minimization Problems: Orthogonal Complements, Minimization Problems.

第六章 内积空间 (6 小时)

6.A 内积与范数

6.B 规范正交基

6.C 正交补与极小化问题

Chapter 7 Operators on Inner Product Spaces (8 hours)

7.A Self-Adjoint and Normal Operators: Adjoints, Self-Adjoint Operators, Normal Operators.

7.B The Spectral Theorem: The Complex Spectral Theorem, The Real Spectral Theorem.

7.C Positive Operators and Isometries: Positive Operators, Isometries.

7.D Polar Decomposition and Singular Value Decomposition: Polar Decomposition, Singular Value Decomposition.

第七章 内积空间上的算子 (8 小时)

7.A 自伴算子与正规算子

7.B 谱定理

7.C 正算子与等距同构

7.D 极分解与奇异值分解

Chapter 8 Operators on Complex Vector Spaces (8 hours)

8.A Generalized Eigenvectors and Nilpotent Operators: Null Spaces of Powers of an Operator, Generalized Eigenvectors, Nilpotent Operators.

8.B Decomposition of an Operator: Description of Operators on Complex Vector Spaces, Multiplicity of an Eigenvalue, Block Diagonal Matrices, Square Roots.

8.C Characteristic and Minimal Polynomials: The Cayley-Hamilton Theorem, The Minimal Polynomial.

8.D The Jordan Form.

第八章 复向量空间上的算子 (8 小时)

8.A 广义本征向量和幂零算子

8.B 算子的分解

8.C 特征多项式和极小多项式

8.D 若尔当形

Chapter 9 Operators on Real Vector Spaces (4 hours)

9.A Complexification: Complexification of a Vector Space, Complexification of an Operator, The Minimal Polynomial of the Complexification, Eigenvalues of the Complexification, Characteristic polynomial of the Complexification.

9.B Operators on Real Inner Product Spaces

Normal Operators on Real Inner Product Spaces, Isometries on Real Inner Product Spaces.

第九章 实向量空间上的算子 (4 小时)

9.A 复化

9.B 实内积空间上的算子

Chapter 10 Trace and Determinant (4 hours)

10.A Trace: Change of Basis: Trace: A Connection Between Operators and Matrices.

10.B Determinant: Determinant of an Operator, Determinant of a Matrix, The Sign of the Determinant, Volume.

第十章 迹与行列式 (4 小时)

10.A 迹

10.B 行列式